

Appendix C — Test Plan Template (UPPMS)

This appendix follows the guide’s Appendix C model, adapted to UPPMS.

C-1 Test Scope

- **In scope:** backend units, integration, API, isolation, scheduling performance, security.
- **Out of full scope:** exhaustive manual UI and all-browser compatibility.

C-2 Test Strategy

Table C-1. Test Strategy

Test Type	Objective	Tool
Unit	Each function/unit separately	PHPUnit Unit
Integration	Units with DB	PHPUnit + Transactions
System	Full API flows	PHPUnit Feature
Performance	GA under denser data	GeneticScheduler + Demo
Security	Isolation, RBAC, XML	Feature + Middleware

C-3 Test Environment

Table C-2. Test Environment

Item	Specification
OS	Windows 10/11
Server	localhost — Laravel
Database	MySQL (test)
UI	React SPA (Vite)
Browsers	Chrome / Edge

C-4 Test Cases (Sample)

Table C-3. Sample Test Cases

ID	Description	Inputs	Expected	Result
TC-01	Matching student registration	Valid XML email+number	active	Done
TC-02	Non-matching student registration	Unknown credentials	Reject	Done
TC-03	Matching supervisor registration	Matching email	active	Done
TC-04	Submit proposal	Title+description+supervisor	pending	Done
TC-05	Approve proposal	Supervisor approve	Project created	Done
TC-06	Block stage skip	Later stage without prior pass	Reject/lock	Done
TC-07	Generate schedule	Projects+availability+rooms	Candidates	Done
TC-08	Committee conflict	Supervisor on own committee	Reject	Done